

Root Words Cheat Sheet V — Astrophysics, Cosmology, Nuclear & Field Theory

This sheet balances the two languages more evenly than the others. Particle physics leans almost entirely Greek (hadron, lepton, meson), so the Latin layer below is worth learning deliberately rather than picking up by accident.

The Mass Hierarchy — HADRON/BARYON vs LEPTON, the load-bearing pair

hadron (Greek *hadros*, "thick, heavy") and **baryon** (Greek *barys*, "heavy"): particles built from quarks, generally the heavy end of the particle zoo.

lepton (Greek *leptos*, "light, small, fine"): particles like the electron and neutrino, generally light and structureless.

meson (Greek *mesos*, "middle"): historically named for sitting in mass between the two.

The mass hierarchy is not a coincidence of naming; it is written directly into the Greek roots.

FUS-/FISS- the second load-bearing pair (Latin *fundere*, to pour / *findere*, to split)

fusion: light nuclei poured together into a heavier one, releasing energy.

fission: a heavy nucleus split apart into lighter fragments, releasing energy.

Same energy-releasing idea, opposite direction, and the roots (pour together vs split apart) tell you which is which without any further explanation.

POTENTI- *power, ability* (Latin *potentia*)

potential, potential energy, potent

NOV- *new* (Latin *novus*)

nova, supernova, novel, innovate

GRADI-/GRAD- *step* (Latin *gradus*)

gradient, degree, graduate, progress, regress

STELL- *star* (Latin *stella*)

stellar, interstellar, constellation

INTENS- *stretch toward* (Latin *intendere*)

intensity, intense

NUCLE- *kernel, nut* (Latin *nucleus*)

nucleus, nuclear, nucleon, nucleosynthesis

SPECTR- *appearance, image* (Latin *specere*, to look)

spectrum, spectroscope, spectral, spectator, aspect, inspect, respect

GRAVIT- *heavy* (Latin *gravis*)

gravity, gravitational, grave — the Latin "heavy" root, standing next to the Greek "heavy" roots (hadron, bary-) above

LUMIN-/LUX *light* (Latin *lumen, lux*)

luminosity, luminous — the Latin "light" root, standing next to lepton (Greek "light") and photo- (Greek "light," sheet four); three languages of physics, one meaning

SINGULAR- *alone, single* (Latin *singularis*)

singularity

ANNIHIL- *to nothing* (Latin *ad- + nihil*)

annihilation — particle-antiparticle pairs converting entirely "to nothing" but energy

QUANT- *how much* (Latin *quantus*)

quantum, quantity, quantise

ELECTRO- *amber* (Greek *elektron*)

electron, electromagnetic — named because static charge was first noticed by rubbing amber

COSMO- *order, world* (Greek *kosmos*)

cosmology, cosmos, cosmic, microcosm

ASTRO-/ASTER- *star* (Greek *astron*)

astronomy, asteroid, astrophysics, disaster (*dis- + astro*, "ill-starred")

GALAXY *milk* (Greek *gala*)

galaxy — from the Milky Way's appearance, same root as "galactic" and the biological "lactose"

PARA- + -ALLAX *alternation* (Greek *parallaxis*)

parallax — literally "an alternation," describing the apparent shift of a star against the background as the observer's position changes

ECCENTR- *out of centre* (Greek *ek- + kentron*)

eccentricity, eccentric — an orbit's eccentricity measures how far its centre is displaced from circular

APO-/PERI- *away from / near* (Greek)

apogee/perigee (Earth), aphelion/perihelion (Sun, sheet four), apoapsis/periapsis (general) — same pair, different central body each time

CAMPUS *field* (Latin *campus* → French *champ*)

field theory / champ de force — English "field" is unrelated Germanic vocabulary, so English and French name the same physics concept from two entirely different word-families; a genuine linguistic coincidence rather than a shared root

Not Every Name Is a Root — BOSON and Friends

boson is named after physicist Satyendra Nath Bose, a proper noun, not a Latin or Greek fossil. **Redshift**, **half-life**, and **event horizon** are all plain, precise English rather than inherited vocabulary. Root-reasoning is a powerful tool, but it stops working the moment a word is a 20th-century coinage rather than an ancient one — knowing which is which matters as much as knowing the roots themselves.

Test Yourself

1. Explain how the roots of *hadron*, *baryon*, and *lepton* directly encode the particle mass hierarchy, without needing any further definition.
2. State the two roots behind *fusion* and *fission* and explain why "pour together" and "split apart" are accurate descriptions of each process.
3. *Gravity* (Latin, "heavy") and *hadron/baryon* (Greek, "heavy") both mean "heavy," but come from different languages. Name one other pair on this or an earlier sheet where two different-language roots converge on the same meaning.
4. *Annihilation* comes from *ad-* + *nihil*, "to nothing." Explain why this is a precise description of what happens in a particle-antiparticle collision.
5. Explain why *electron* is named after amber, and what this tells you about how the ancient Greeks first encountered the phenomenon of static charge.
6. *Galaxy* shares a root with the everyday word for milk. Explain the connection.
7. *Parallax* literally means "an alternation." Explain how this describes the method astronomers use to measure the distance to nearby stars.
8. English "field" and French "champ" name the same physics concept but come from entirely unrelated word-families. Explain why this matters when you're trying to guess a word's meaning from its roots in a foreign-language physics text.

9. *Boson* is not built from a Latin or Greek root at all. Explain where the name actually comes from, and why this is worth knowing before you try to "decode" it etymologically.

10. *Apogee*, *aphelion*, and *apoapsis* all share the same prefix but refer to different central bodies. State the prefix's meaning and explain what distinguishes the three words from each other.

Answer Bank

1. *Hadron* and *baryon* both come from Greek roots meaning "heavy" (*hadros*, *barys*), while *lepton* comes from *leptos*, "light, small." The words themselves state the mass hierarchy directly: hadrons and baryons are the heavy particles, leptons are the light ones, with mesons (*mesos*, "middle") sitting between.
2. *Fusion* comes from *fundere*, "to pour," describing light nuclei being poured or merged together into a heavier one. *Fission* comes from *findere*, "to split," describing a heavy nucleus being split apart into lighter fragments. Both release energy, but the roots tell you immediately which direction each process runs.
3. Any valid example is acceptable, for instance: *photo-* (Greek, light) and *lumin-/lux* (Latin, light) both mean light; or *iso-* (Greek) and *equi-* (Latin) both mean equal. The point is that physics vocabulary draws on both languages for the same underlying ideas.
4. In annihilation, a particle and its antiparticle collide and convert entirely into energy (typically photons); no rest mass survives as matter, so the particles are reduced "to nothing" in the literal Latin sense, even though the energy itself is conserved.
5. The ancient Greeks discovered that rubbing amber (Greek *elektron*) against cloth caused it to attract light objects, an early observation of static electricity. When the phenomenon was formally named centuries later, the Greek word for amber was reused to name both electricity and, eventually, the electron itself.
6. *Galaxy* comes from Greek *gala*, "milk," because the Milky Way appears as a pale, milky band of light across the night sky. The same root gives "lactose" and "lactation" in biology, an entirely different field describing the same underlying image of milk.
7. Parallax measures the apparent shift, or "alternation," in a star's position against distant background stars when viewed from two different points in Earth's orbit six months apart. That apparent shift is used, via simple trigonometry, to calculate the star's distance.
8. Etymology only works within a language family; a root that looks unrelated in another language (like "field" vs *champ*) can still name the identical physical concept. Two words can converge on the same meaning by coincidence of history rather than shared ancestry, so root-reasoning has to be treated as a strong hint, not a guarantee, when crossing between languages.
9. *Boson* is named after the Indian physicist Satyendra Nath Bose, who developed the statistics that describe this class of particle. It carries no etymological meaning at all beyond being a surname, so trying to "decode" it via Latin or Greek roots would be a wasted effort; some technical vocabulary simply has to be learned as a fact, not derived.
10. The shared prefix is *apo-*, "away from," always naming the point in an orbit furthest from the central body. The three words differ only in which central body they refer to: *-gee* refers to Earth (Greek *ge*, earth), *-helion* refers to the Sun (Greek *helios*), and *-apsis* is the general term used regardless of the central body.

Matching Quiz: Prefix to Meaning

Draw a line (or write the matching letter) connecting each prefix/root on the left to its correct meaning on the right. Meanings are deliberately shuffled and there are more meanings than prefixes, so some will be left over.

Prefix/Root		Meaning
1. HADRON-/BARY-	_____	A. light, small
2. LEPTON-	_____	B. heavy
3. MESON-	_____	C. new
4. NOV-	_____	D. middle
5. GRAVIT-	_____	E. to nothing
6. SPECTR-	_____	F. heavy (Latin)
7. ANNIHIL-	_____	G. appearance, to look
8. COSMO-	_____	H. star
9. ASTRO-	_____	I. order, world
10. QUANT-	_____	J. how much
		K. alone, single
		L. step

Bonus: pick one leftover meaning from the right-hand column and name a word from any of the five sheets that uses it.

Tricky Words: Construct Your Own Definition

1. **Isotropic** (*iso-* = equal, *tropos* = turning) — *callback, worth re-testing in this context*
Your definition:

2. **Nucleosynthesis** (*nucleus* = kernel, *syn-* = together, *thesis* = a placing)
Your definition:

3. **Degenerate (matter)** (*de-* = down, away, *genus* = kind, birth)
Your definition:

4. **Quintessence** (*quinta* = fifth, *essentia* = essence, being)
Your definition:

5. **Inflation** (*in-* = into, *flare* = to blow)
Your definition:

6. **Recombination** (*re-* = again, *com-* = together, *binare* = to join two by two)
Your definition: